

Prompt Engineering

Final Examination

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Instructions:

- This exam consists of 40 multiple-choice questions.
 - Each question is worth 2.5 points for a total of 100 points.
 - Select the **best** answer for each question.
 - You have 90 minutes to complete this exam.
 - No electronic devices or notes are permitted.
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1. A **prompt** in the context of AI is best defined as:
 - A) The internal memory of an LLM
 - B) The training data used to build the model
 - C) The hardware that runs the AI system
 - D) The input message you give to an AI system to elicit a response
2. Which of the following is NOT one of the core elements that good prompts provide?
 - A) Purpose
 - B) Context
 - C) Constraints
 - D) Source code
3. What is **tokenization** in the context of LLMs?
 - A) The process of encrypting user data
 - B) Generating authentication tokens for API access
 - C) Converting images into text descriptions
 - D) Breaking text into smaller chunks (tokens) that the model can process
4. Modern LLMs like GPT-4 use subword tokenization primarily because:
 - A) It makes the model run faster
 - B) It reduces the cost of training
 - C) It improves image recognition
 - D) It can handle new or rare words by combining familiar pieces

5. What are **embeddings** in LLM architecture?
 - A) Physical components of the computer running the model
 - B) The user interface elements of AI applications
 - C) Error logs generated during model training
 - D) Numerical representations of tokens that capture semantic meaning
6. The 2017 paper “Attention Is All You Need” introduced which key innovation?
 - A) Self-attention mechanism (Transformer architecture)
 - B) Tokenization
 - C) Recurrent neural networks
 - D) Convolutional neural networks
7. **Hallucination** in LLMs refers to:
 - A) The model refusing to answer questions
 - B) The model asking clarifying questions
 - C) The model generating plausible but incorrect or fabricated information
 - D) The model running out of memory
8. Which statement about LLM capabilities is TRUE?
 - A) LLMs have true understanding and consciousness
 - B) LLMs have unlimited context windows
 - C) LLMs are statistical pattern learners, not conscious reasoners
 - D) LLMs never produce biased outputs
9. In the anatomy of a good prompt, which element specifies headings, JSON schema, or table columns?
 - A) Task
 - B) Context
 - C) Constraints
 - D) Output Specification
10. **Zero-shot prompting** is best described as:
 - A) Giving just the task without any examples
 - B) Providing multiple examples before the task
 - C) Using images along with text
 - D) Chaining multiple prompts together
11. **Few-shot prompting** is most useful when:
 - A) You want to save tokens
 - B) The task is extremely simple
 - C) You need to steer style or format with specific examples
 - D) You want to reduce response time
12. What is **role prompting**?

- A) Asking the model to play a video game
 - B) Defining database roles
 - C) Setting a persona, audience, and tone for the model's response
 - D) Assigning user permissions to the model
13. Which is an example of a "prompt anti-pattern"?
- A) Specifying output format clearly
 - B) Providing relevant context
 - C) Setting clear constraints
 - D) Including conflicting directives like "be brief" and "include all details"
14. In Retrieval-Augmented Generation (RAG), what is the primary purpose?
- A) To speed up model inference
 - B) To reduce the model size
 - C) To ground answers with provided excerpts or references
 - D) To train the model on new data
15. The **CO-STAR** framework stands for:
- A) Code, Output, Style, Testing, Analysis, Review
 - B) Context, Objective, Style, Tone, Audience, Response
 - C) Constraint, Optimization, Structure, Target, Action, Result
 - D) Chain, Order, Sequence, Task, Answer, Repeat
16. The **IDEA Loop** framework for prompt iteration consists of:
- A) Implement, Debug, Edit, Apply
 - B) Input, Design, Execute, Analyze
 - C) Initiate, Develop, Enhance, Archive
 - D) Identify, Draft, Evaluate, Adjust
17. **Meta-prompting** refers to:
- A) Using prompts to have an LLM generate or refine other prompts
 - B) Prompting about metadata
 - C) Using multiple models simultaneously
 - D) Prompting in multiple languages
18. In the context of tool use and function calling, the LLM typically:
- A) Directly executes code on the user's computer
 - B) Sends requests to arbitrary websites
 - C) Generates a structured object (e.g., JSON) specifying the tool and arguments
 - D) Modifies its own training data
19. Which dimension of linguistic register controls terminology density?
- A) Tenor (Relationship)

- B) Mode (Channel)
 - C) Field (Topic Domain)
 - D) Rhythm
20. “CEFR B1” in prompt specifications refers to:
- A) A programming language version
 - B) A model architecture type
 - C) A security classification
 - D) A reading/language proficiency level
21. **Brand Voice Fingerprinting** involves:
- A) Creating biometric authentication
 - B) Recording audio samples of brand representatives
 - C) Extracting measurable voice attributes from existing content to ensure AI outputs match brand style
 - D) Generating unique identifiers for users
22. The **Critique Loop** in prompt refinement typically follows which sequence?
- A) Generate → Deploy → Monitor
 - B) Design → Test → Archive
 - C) Plan → Execute → Report
 - D) Generate → Critique → Revise
23. “Vibe Coding” refers to:
- A) Writing code by describing desired behavior to an LLM and iterating based on output’s look and feel
 - B) Writing code based on musical rhythms
 - C) A formal software development methodology
 - D) Debugging code using vibration sensors
24. What is the main advantage of using “LLM-as-a-Judge” for evaluation?
- A) It is always 100% accurate
 - B) It eliminates the need for any human review
 - C) It enables scalable evaluation of thousands of outputs quickly
 - D) It reduces model training costs
25. The “3 H’s” framework for AI output evaluation includes Helpful, Honest, and:
- A) Historical
 - B) Hypothetical
 - C) Harmless
 - D) Hierarchical
26. A **Golden Dataset** in AI evaluation is:
- A) A curated set of inputs with “perfect” outputs used as benchmarks

- B) The largest available training dataset
 - C) Data stored in a premium database
 - D) User feedback collected over time
27. **Verbosity Bias** in LLM evaluation refers to:
- A) Models producing too few tokens
 - B) Preference for technical jargon
 - C) The tendency to rate longer answers as better, even if they contain fluff
 - D) Bias toward English-language responses
28. In Vision-Language Models (VLMs), images are converted into:
- A) Audio signals
 - B) Database entries
 - C) Encrypted files
 - D) Patches that become “visual tokens”
29. **Set-of-Mark (SoM) Prompting** involves:
- A) Overlaying numbered tags or bounding boxes on images to improve grounding
 - B) Grading student assignments
 - C) Creating watermarks on AI-generated images
 - D) Setting bookmarks in long documents
30. What is **Visual Hallucination**?
- A) The model refusing to process images
 - B) Image compression artifacts
 - C) The model describing objects or details that aren’t present in the image
 - D) Creating optical illusions
31. The **MECE Principle** stands for:
- A) Mutually Exclusive, Collectively Exhaustive
 - B) Maximum Efficiency, Controlled Execution
 - C) Model Evaluation, Context Enhancement
 - D) Multiple Examples, Chain Execution
32. **Tree of Thoughts (ToT)** differs from Chain-of-Thought by:
- A) Using fewer tokens
 - B) Being faster to execute
 - C) Requiring no examples
 - D) Exploring multiple reasoning branches and allowing backtracking
33. The **ReAct** framework combines:
- A) Recalling and Testing
 - B) Reading and Acting

- C) Reasoning and Acting
 - D) Responding and Chatting
34. In **Self-Consistency** prompting, you:
- A) Ask the model to repeat the same answer multiple times
 - B) Force the model to be consistent with previous conversations
 - C) Use the same prompt for different models
 - D) Generate multiple independent answers using different reasoning paths and pick the most consistent
35. The **SOAP Note** format (Subjective, Objective, Assessment, Plan) is commonly used in which domain?
- A) Legal
 - B) Finance
 - C) Healthcare
 - D) Engineering
36. The **IRAC** framework (Issue, Rule, Analysis, Conclusion) is specifically designed for:
- A) Medical diagnosis
 - B) Financial reporting
 - C) Software development
 - D) Legal reasoning and analysis
37. A **Knowledge Pack** in domain-specific prompting typically includes:
- A) Definitions, acronyms, and style guides specific to the domain
 - B) Only training data
 - C) Password credentials
 - D) Model weights
38. The term **Automation Bias** refers to:
- A) Bias in robotic systems
 - B) Preference for automated testing
 - C) The tendency for humans to over-rely on AI outputs without critical verification
 - D) AI systems preferring automated tasks
39. What is **Prompt Injection**?
- A) Adding more tokens to a prompt
 - B) Injecting code into the model's training
 - C) A medical procedure
 - D) Crafting inputs to deliberately bypass AI safety protocols
40. **Agentic AI** differs from traditional chatbots by being:
- A) Unable to use external tools
 - B) Text-only without multimodal capabilities

- C) Proactive, multi-step, stateful, and capable of autonomous tool usage
 - D) Slower and less accurate
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— END OF EXAM —

Answer Key (For Instructor Use Only)

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	C	31	A
2	D	12	C	22	D	32	D
3	D	13	D	23	A	33	C
4	D	14	C	24	C	34	D
5	D	15	B	25	C	35	C
6	A	16	D	26	A	36	D
7	C	17	A	27	C	37	A
8	C	18	C	28	D	38	C
9	D	19	C	29	A	39	D
10	A	20	D	30	C	40	C

Detailed Answer Key with Explanations

1. **D** - A prompt is the input message given to an AI system.
2. **D** - Source code is not a core element; purpose, context, and constraints are.
3. **D** - Tokenization breaks text into processable chunks.
4. **D** - Subword tokenization handles new/rare words flexibly.
5. **D** - Embeddings are numerical vectors capturing semantic meaning.
6. **A** - The Transformer architecture introduced self-attention.
7. **C** - Hallucination is generating plausible but incorrect information.
8. **C** - LLMs are pattern learners, not conscious reasoners.
9. **D** - Output Specification defines format like JSON, tables.
10. **A** - Zero-shot means no examples, just the task.
11. **C** - Few-shot steers style/format with examples.
12. **C** - Role prompting sets persona, audience, tone.
13. **D** - Conflicting directives are a common anti-pattern.
14. **C** - RAG grounds answers with provided context.
15. **B** - CO-STAR: Context, Objective, Style, Tone, Audience, Response.
16. **D** - IDEA Loop: Identify, Draft, Evaluate, Adjust.
17. **A** - Meta-prompting uses prompts to generate/refine prompts.
18. **C** - Tool use generates structured JSON for function calls.
19. **C** - Field (Topic Domain) controls terminology density.
20. **D** - CEFR B1 is a language proficiency level.

21. **C** - Brand Voice Fingerprinting extracts measurable attributes.
22. **D** - Critique Loop: Generate → Critique → Revise.
23. **A** - Vibe Coding is iterating on LLM output based on feel.
24. **C** - LLM-as-Judge enables scalable evaluation.
25. **C** - The 3 H's: Helpful, Honest, Harmless.
26. **A** - Golden Dataset is a curated benchmark set.
27. **C** - Verbosity Bias favors longer answers.
28. **D** - VLMs convert images into visual tokens.
29. **A** - Set-of-Mark overlays numbered markers on images.
30. **C** - Visual Hallucination describes non-existent details.
31. **A** - MECE: Mutually Exclusive, Collectively Exhaustive.
32. **D** - ToT explores multiple branches with backtracking.
33. **C** - ReAct combines Reasoning and Acting.
34. **D** - Self-Consistency generates multiple answers to find consensus.
35. **C** - SOAP Notes are used in Healthcare.
36. **D** - IRAC is for Legal reasoning.
37. **A** - Knowledge Packs include definitions, acronyms, style guides.
38. **C** - Automation Bias is over-reliance on AI outputs.
39. **D** - Prompt Injection bypasses AI safety protocols.
40. **C** - Agentic AI is proactive, multi-step, stateful.

Topic Coverage by Question

- Foundations: Q1–Q8
- Prompt Design Principles: Q9–Q14
- Patterns and Frameworks: Q15–Q18
- Style, Tone, and Format Control: Q19–Q21
- Iterative Refinement: Q22–Q24
- Evaluation Methods: Q25–Q27
- Multimodal Prompting: Q28–Q30
- Complex Problem Solving: Q31–Q34
- Domain-Specific Applications: Q35–Q37
- Ethics, Responsible AI & Emerging Trends: Q38–Q40